

What Makes a Bestseller?

Analysis of Amazon Top 50 Bestselling Books, 2009–2019

Data Analytics Case Study

In this project, I explored Amazon's Top 50 bestselling books from 2009 to 2019. I looked at genre, ratings, prices, and authors to see what patterns appeared in the dataset.

The goal was to understand what characteristics were common among books on the bestseller list.

Tools:

- **Python & Pandas** - cleaning and initial exploration
- **SQL / BigQuery** - analysis and calculations
- **Tableau** - visualization

1. Ask

Business Task

The goal of this analysis is to help a book retailer understand what bestselling books in this dataset have in common. The analysis could help with decisions about product selection and marketing.

Business Questions

1. Which genre is most common among bestselling books?
2. How do average ratings differ between Fiction and Non Fiction books?
3. How does the average price differ between the two genres?
4. Which authors appear most often on the bestseller list?
5. How does the distribution of Fiction and Non Fiction books change over time?

2. Prepare

Data Source

For this project, I used the Amazon Top 50 Bestselling Books 2009–2019 dataset from Kaggle. It contains information about books that appeared in Amazon's Top 50 bestseller list from 2009 to 2019.

Dataset Overview

The dataset contains **550 rows and 7 columns**.

The main columns include:

- **Book Name** - name of the book
- **Author** - author of the book
- **User Rating** - average user rating
- **Reviews** - number of reviews
- **Price** - book price
- **Year** - year the book appeared on the list
- **Genre** - Fiction or Non Fiction

Data Quality

Data quality checks

I checked:

- number of rows and columns
- data types
- missing values
- duplicate rows

After these checks, I found no missing values or duplicate rows.

```
df.info()

<class 'pandas.DataFrame'>
RangeIndex: 550 entries, 0 to 549
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Name             550 non-null   str
1   Author           550 non-null   str
2   User Rating      550 non-null   float64
3   Reviews          550 non-null   int64
4   Price            550 non-null   int64
5   Year             550 non-null   int64
6   Genre            550 non-null   str
dtypes: float64(1), int64(3), str(3)
memory usage: 30.2 KB
```

```
df.isnull().sum()

Name          0
Author        0
User Rating   0
Reviews       0
Price         0
Year          0
Genre         0
dtype: int64
```

```
df.duplicated().sum()

np.int64(0)
```

Data Limitations

There are a few limitations to this dataset. It only includes the Top 50 books for each year, so it does not represent all books sold on Amazon. It also does not contain actual sales or revenue data, customer demographics, or information about individual purchases.

For this project, the dataset was enough to compare Fiction and Non Fiction, but it cannot be used to describe the entire Amazon book market.

3. Process

Tools

I used Python and Pandas to clean and explore the dataset. I used SQL in Google BigQuery to calculate summary statistics and answer the business questions.

Data Cleaning

After checking the data, I used Python and Pandas to prepare it for analysis. I renamed the columns to make them easier to work with in Python and SQL.

```
df = df.rename(columns={
    'Name': 'book_name',
    'Author': 'author',
    'User Rating': 'user_rating',
    'Reviews': 'reviews',
    'Price': 'price',
    'Year': 'year',
    'Genre': 'genre'
})
```

```
df.head()
```

	book_name	author	user_rating	reviews	price	year	genre
0	10-Day Green Smoothie Cleanse	JJ Smith	4.7	17350	8	2016	Non Fiction
1	11/22/63: A Novel	Stephen King	4.6	2052	22	2011	Fiction
2	12 Rules for Life: An Antidote to Chaos	Jordan B. Peterson	4.7	18979	15	2018	Non Fiction
3	1984 (Signet Classics)	George Orwell	4.7	21424	6	2017	Fiction
4	5,000 Awesome Facts (About Everything!) (Natio...	National Geographic Kids	4.8	7665	12	2019	Non Fiction

Renaming columns and checking the cleaned dataset in Python

SQL Preparation

I uploaded the cleaned dataset to Google BigQuery and created a table called **books**. I then used SQL to calculate summary statistics and answer the business questions.

```
1 SELECT
2   genre,
3   ROUND(AVG(price), 2) AS average_price
4 FROM `glassy-emblem-476516-r3.amazon_books.books`
5 GROUP BY genre
6 ORDER BY average_price DESC;
7
```

✓ This script will process 31.03 KB when run.

>I Query results

Job information		Results	Visualization	JSON	Execution details
Row	genre	average_price			
1	Non Fiction	14.84			
2	Fiction	10.85			

4. Analyze

Key Findings

The analysis showed several differences between Fiction and Non Fiction books on the bestseller list.

1. Non Fiction books were more common

There were **310 Non Fiction** books compared with **240 Fiction books**. Non Fiction books made up about **56%** of the dataset, while Fiction books made up about **44%**.

2. Fiction books had a slightly higher average rating

Fiction books had an average rating of **4.65**, compared with **4.60** for Non Fiction books. The difference was small, so the average ratings were very similar.

The screenshot shows a SQL query editor with the following code:

```

8 SELECT
9   genre,
10  ROUND(AVG(user_rating), 2) AS average_rating
11 FROM `glassy-emblem-476516-r3.amazon_books.books`
12 GROUP BY genre
13 ORDER BY average_rating DESC;

```

Below the query, a green checkmark indicates: "This script will process 20.68 KB when run." and "Using on-demand processing quota".

The "Query results" section shows a table with two columns: "genre" and "average_rating".

Row	genre	average_rating
1	Fiction	4.65
2	Non Fiction	4.6

3. Non Fiction books had a higher average price

The average price of a Non Fiction book was **\$14.84**, compared with **\$10.85** for Fiction books. This shows a noticeable difference in the average price between the two genres. Non Fiction books were about **\$3.99 more expensive on average**.

4. Some authors appeared on the bestseller list more often

Jeff Kinney appeared most frequently in the dataset, with **12 books**. Other frequently appearing authors included **Suzanne Collins**, **Rick Riordan**, and **Gary Chapman**.

5. Genre distribution changed over time

The genre distribution was not the same every year. Non Fiction appeared more often in most years, although the gap between the two genres changed over time.

The trend is shown in the Tableau visualization on the next page.

5. Share

Tableau Dashboard

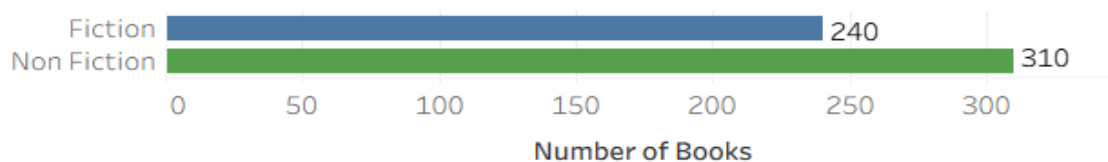
I created a Tableau dashboard to visualize the main findings from the analysis. The dashboard includes four visualizations:

- **Books by Genre** - shows the number of Fiction and Non Fiction books in the dataset.
- **Average Rating by Genre** - compares the average rating of Fiction and Non Fiction books.
- **Average Price by Genre** - compares the average price of books in each genre.
- **Books by Genre Over Time** - shows how the number of Fiction and Non Fiction books changed from 2009 to 2019.

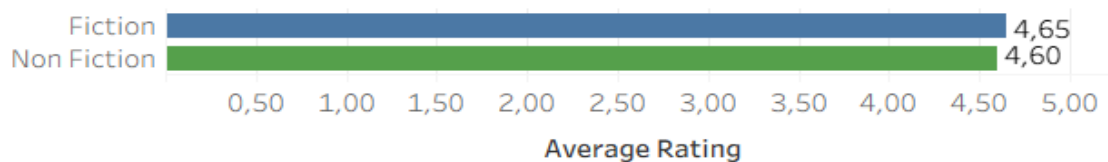
What Makes a Bestseller?

Amazon Top 50 Bestselling Books, 2009–2019

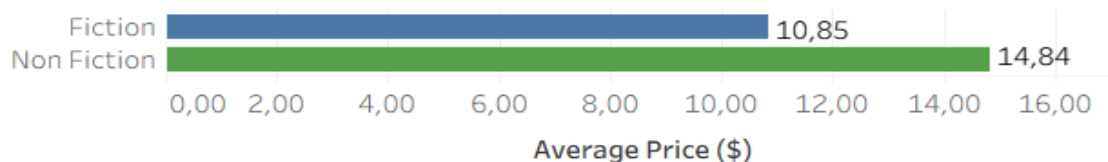
Books by Genre



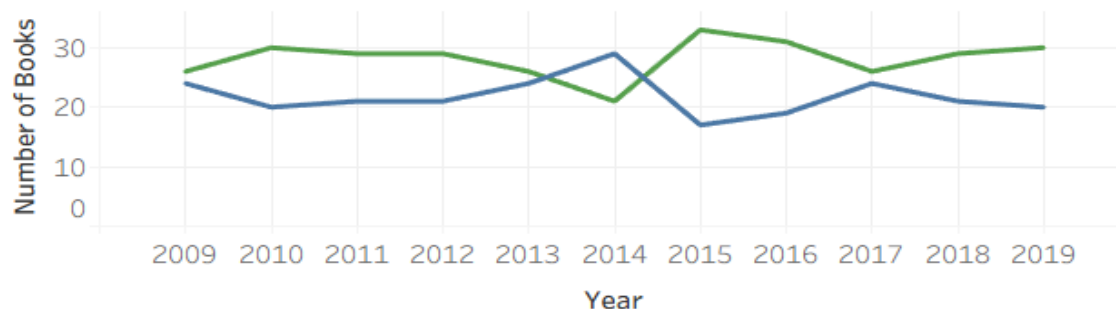
Average Rating by Genre



Average Price by Genre



Books by Genre Over Time



Key takeaway

Non Fiction books appeared more often and had a higher average price, while Fiction books had a slightly higher average rating.

6. Act

Final Conclusion

Based on this dataset, Non Fiction books appeared more often on the bestseller list than Fiction books. There were 310 Non Fiction books compared with 240 Fiction books.

Non Fiction books also had a higher average price, at \$14.84 compared with \$10.85 for Fiction books. Fiction books had a slightly higher average rating, but the difference was small.

Overall, the data shows some differences between Fiction and Non Fiction books. However, sales and customer data would be needed to understand what really drives book sales.

Top 3 Recommendations

1. Maintain a strong Non Fiction selection

The retailer could maintain a strong Non Fiction selection because it appeared more often in the bestseller dataset.

2. Promote highly rated Fiction books

The retailer could consider promoting highly rated Fiction books because Fiction had a slightly higher average rating.

3. Consider genre when reviewing prices

The retailer could consider genre when reviewing book prices, as Non Fiction books had a higher average price.